

JAYESH CHOTALIYA

Bangalore, India — chotaliyajayesh3@gmail.com — +91 97144 47996 — linkedin.com/in/jayeshchotaliya

PROFESSIONAL EXPERIENCE

Associate R&D Engineer - Embedded Software Engineer

September 2023 – Present

ABB Global Industries and Services Pvt. Ltd., Bangalore, India

- Architected secure cloud connectivity for industrial drives using TPM and certificate authentication, enabling predictive maintenance and reducing downtime by 30%
- Engineered firmware modules on LmP OS (Yocto-based) for real-time data exchange across DDCS, Phloem-eth, and OPC UA protocols
- Developed OPC UA server/client modules for standardized industrial equipment data exchange and real-time drive monitoring
- Designed modular architecture with MQTT/ZeroMQ message-based IPC, improving maintainability and scalability
- Created multithreaded data aggregation pipelines for telemetry and events, optimizing memory and reducing bandwidth
- Implemented Azure IoT Hub integration with device provisioning, twin sync, and secure file upload for remote management
- Performed system bring-up, UART/I2C debugging, and collaborated with firmware/hardware teams for integration
- Established test frameworks with Google Test and CMake achieving 80%+ coverage; integrated HIL testing
- Optimized data pipelines achieving 40% throughput improvement and 99.9% uptime through robust error handling

Technologies: C++17, C, Boost, nlohmann_json, LmP OS, Yocto, OPC UA, MQTT, TPM, ZeroMQ, Azure IoT Hub, CMake, Google Test, UART, I2C

Embedded Systems Trainee

March 2023 – September 2023

Vector India Pvt. Ltd., Bangalore, India

- Built bare-metal applications and device drivers on ARM Cortex-M using Embedded C and CMSIS
- Developed RTOS schedulers and ISR-driven applications for GPIO, UART, ADC, and CAN peripherals
- Gained expertise in memory-mapped I/O, register programming, and debugging with Keil MDK, Proteus, GDB
- Worked with Linux cross-compilation toolchains (GCC ARM) and embedded development workflows

Technologies: Embedded C, ARM Cortex-M, CMSIS, FreeRTOS, Keil MDK, Proteus, GDB, UART, I2C, SPI, CAN

TECHNICAL SKILLS

Languages: C, C++ (C++11/14/17, STL, Boost), Python, Shell Scripting, Embedded C

Embedded Systems: ARM Cortex-M, LPC2129, STM32, Bare-metal, RTOS (FreeRTOS), Embedded Linux, Yocto, LmP OS

Protocols & Interfaces: OPC UA, MQTT, ZeroMQ, Modbus, CAN 2.0A, UART, I2C, SPI, GPIO, TCP/IP

Tools & Frameworks: Git, CMake, GCC, Keil MDK, Proteus, GDB, Google Test, Docker, CI/CD, CMSIS, nlohmann_json

Cloud & Security: Azure IoT Hub, Device Provisioning, TPM, Certificate Authentication, Secure Boot, TLS/SSL

PROJECTS

C Compiler – Preprocessor Stage — Vector India Pvt. Ltd.

August 2023

- Designed GCC-compliant preprocessor in C for macro expansion, conditional compilation, and header inclusion
- Optimized lexical analysis achieving 25% reduction in preprocessing time through efficient hash tables

Body Control Module – CAN Protocol (ARM LPC2129) — Vector India Pvt. Ltd.

July 2023

- Implemented BCM using CAN 2.0A on ARM7-based LPC2129 for automotive applications
- Developed interrupt-driven firmware to control vehicle LEDs via CAN with bit timing and bus arbitration

EDUCATION

Bachelor of Engineering in Electronics and Communication — CGPA: 7.92/10

July 2016 – Aug 2020

L.D. College of Engineering, Ahmedabad, India

Higher Secondary (12th) — 74%

March 2016 — St. Xavier's High School, Surat

Secondary (10th) — 81%

March 2014 — J.V. High School, Surat

CERTIFICATIONS & ACTIVITIES

Embedded Systems Training — Vector India Pvt. Ltd.

March 2023 – September 2023

Leadership: ABB Cultural/Food Committee member — 1st prize Indian National Theatre — GTU Tech Fest operations lead

Notable Projects: Contributed to ABB EGW-02 Connectivity Edge Gateway (Red Dot Award-winning product)